Case Study 10: Food Waste Monitoring System

Assignment Case No. 53 | Level: Beginner

01 PROBLEM STATEMENT

Compare prepared meals, meals served, and leftovers across several days; calculate wastage percentages and identify days exceeding acceptable limits.

Problem interpretation: A canteen records meals prepared and meals served each day. The program calculates leftovers and wastage percentage, then flags days where the wastage percentage exceeds the permitted limit.

02 OBJECTIVES

Store daily meal counts in lists.

Calculate leftover meals.

Calculate wastage percentage.

Flag days exceeding the acceptable wastage limit.

03 ASSUMPTIONS

Meals prepared must be greater than zero.

Meals served cannot be negative or exceed meals prepared.

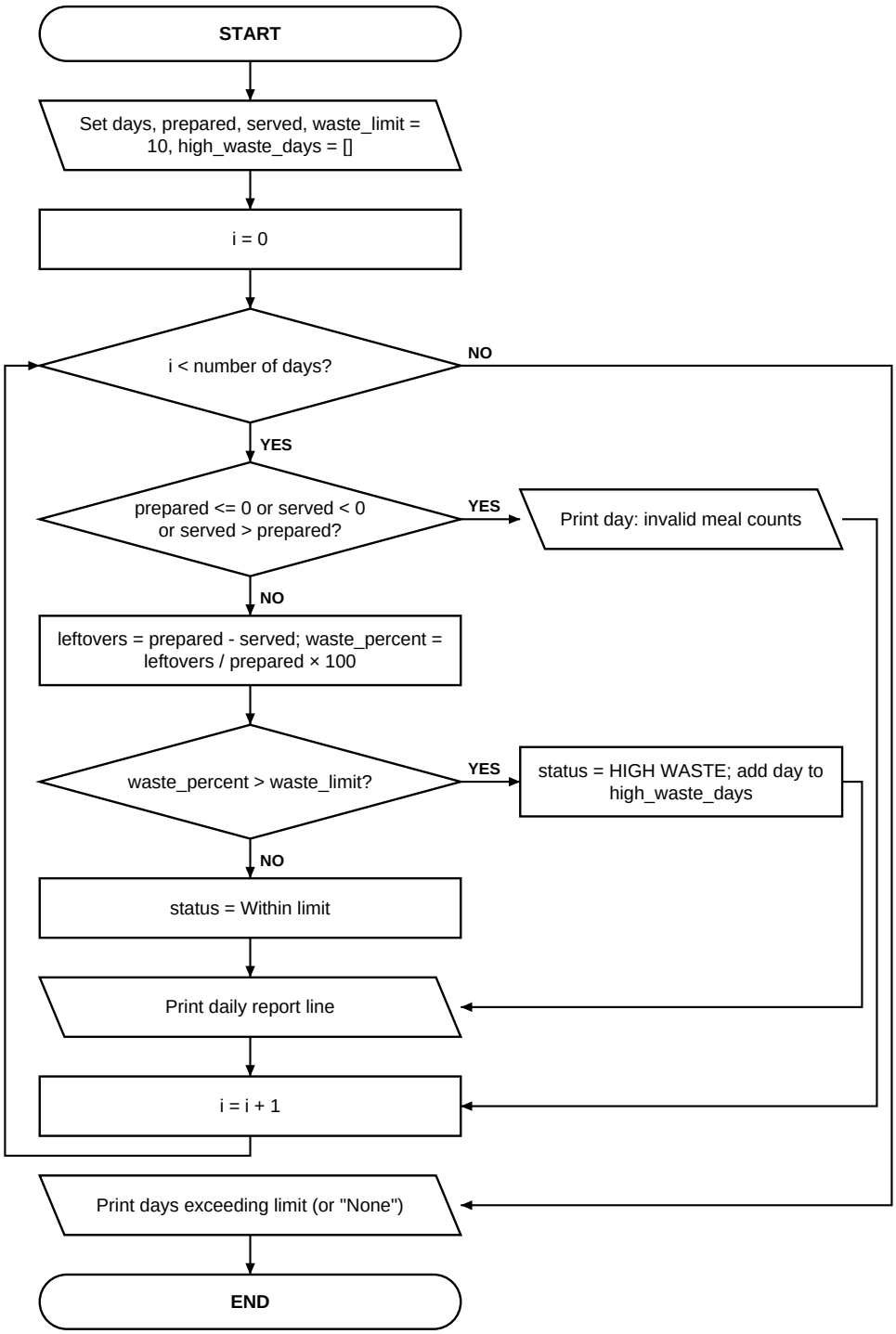
Wastage percentage = leftovers / prepared × 100.

The acceptable wastage limit is 10%.

04 ALGORITHM

1. Start.
2. Store days, meals prepared, and meals served.
3. For each day validate the counts.
4. Calculate leftovers and wastage percentage.
5. Compare the percentage with the 10% limit.
6. Display daily results and flag high-waste days.
7. Stop.

05 FLOWCHART (TEXT FLOW)



06 PSEUDOCODE

```
BEGIN
    days = [Monday, Tuesday, Wednesday, Thursday]
    prepared = [100, 120, 90, 110]
    served = [92, 100, 85, 88]
    waste_limit = 10; high_waste_days = []
    FOR each day i
        IF prepared[i] <= 0 OR served[i] < 0 OR served[i] > prepared[i] THEN
            PRINT days[i], "invalid meal counts"
        ELSE
            leftovers = prepared[i] - served[i]
            waste_percent = leftovers / prepared[i] * 100
            IF waste_percent > waste_limit THEN
                status = "HIGH WASTE"
            ELSE
                status = "Within limit"
            END IF
            PRINT days[i], prepared, served, leftovers, waste_percent, status
            IF waste_percent > waste_limit THEN add days[i] to high_waste_days
        END IF
    END FOR
    PRINT high_waste_days, or "None" if empty
END
```

07 PYTHON SOURCE CODE

The complete Python program is provided in the separate file 10_Food_Waste_Monitoring_System_Code.pdf.

08 TEST CASES

INPUT / SCENARIO	CHECK	EXPECTED RESULT
Prepared 100, served 92	8.00% wastage	Within limit
Prepared 120, served 100	16.67% wastage	HIGH WASTE; day flagged
Prepared 100, served 120	Served exceeds prepared	Invalid meal counts
Prepared 100, served 90	Exactly 10% (boundary)	Within limit; not flagged

09 SAMPLE OUTPUT PREVIEW

```
FOOD WASTE REPORT
Monday: prepared=100, served=92, leftovers=8, waste=8.00% - Within limit
Tuesday: prepared=120, served=100, leftovers=20, waste=16.67% - HIGH WASTE
Wednesday: prepared=90, served=85, leftovers=5, waste=5.56% - Within limit
Thursday: prepared=110, served=88, leftovers=22, waste=20.00% - HIGH WASTE
Days exceeding 10 %: ['Tuesday', 'Thursday']
```

10 CONCLUSION

The program calculates leftovers and waste percentages, applies a threshold, and helps a canteen identify days that need better meal planning.